



Province of the
EASTERN CAPE
EDUCATION

Iphondo leMpuma Kapa: Isabe leMfundo
Provinsie van die Oos Kaap: Departement van Onderwys
Porafensie Ya Kapa Botjahabela: Lefapha la Thuto

**NATIONAL
SENIOR CERTIFICATE/
*NASIONALE
SENIOR SERTIFIKAAT***

GRADE/*GRAAD* 11

NOVEMBER 2025

**MATHEMATICS P2/*WISKUNDE V2*
MARKING GUIDELINE/*NASIENRIGLYN***

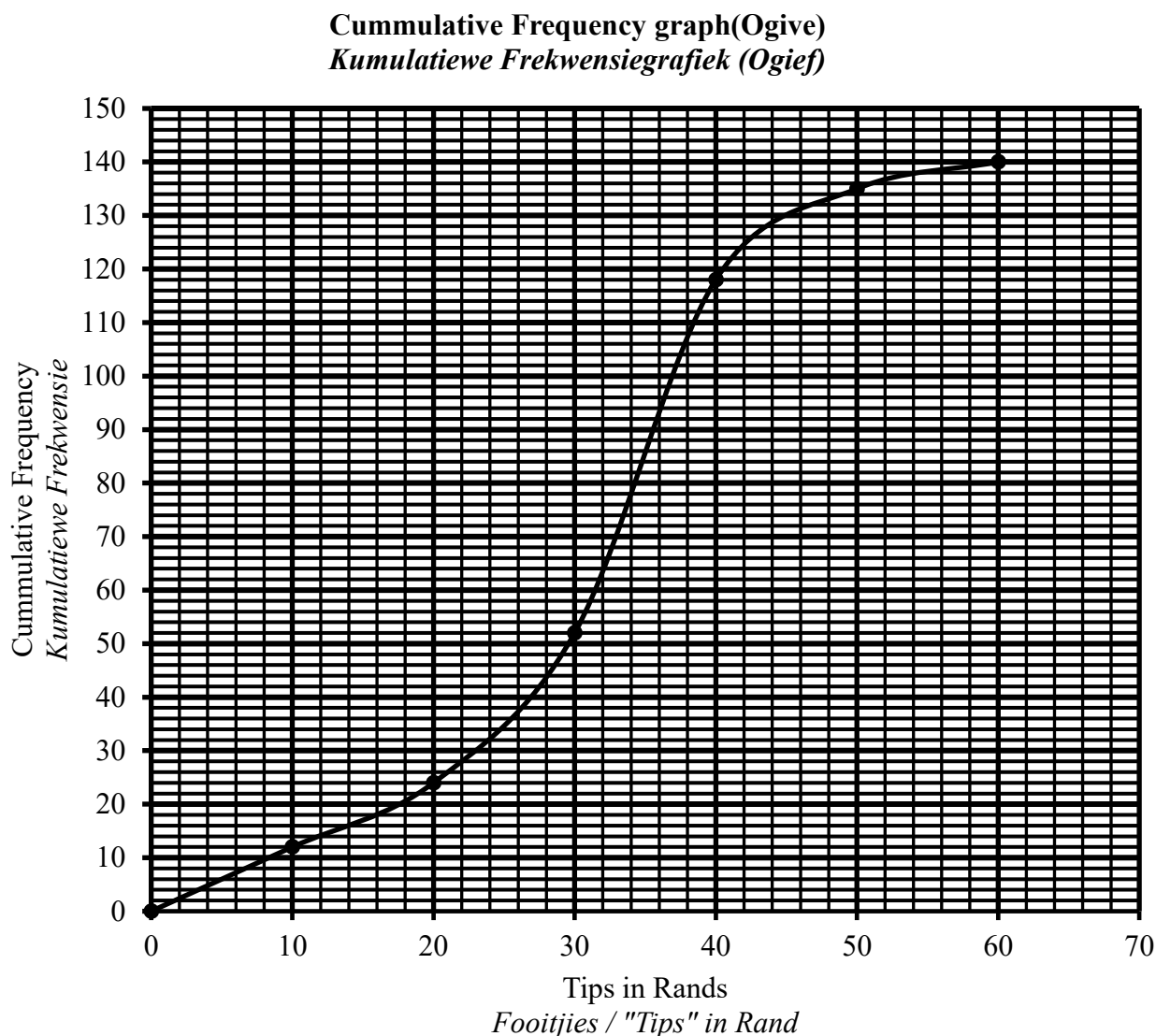
MARKS/*PUNTE*: 150

This marking guideline consists of 16 pages./
Hierdie nasienriglyn bestaan uit 16 bladsye.

QUESTION 1/VRAAG 1

Data Set/ <i>Datastel</i> :										
Prizes/ <i>Pryse</i>	3 600	4 500	5 400	6 200	x	12 300	15 800	y	43 800	87 500
1.1	$\frac{x + 12300}{2} = 9700$ $x = 7100$					✓ method/ <i>metode</i> ✓ answer/ <i>antwoord</i>				(2)
1.2	$\frac{y + 186200}{10} = 21770$ $y = 31500$					✓ correct equation <i>korrekte vergelyking</i> ✓ $y + 186200$ ✓ answer/ <i>antwoord</i>				(3)
1.3	outlier/ <i>uitskieter</i> = 87 500					✓ answer/ <i>antwoord</i>				(1)
1.4	$\delta = 25258,19$					✓✓ answer/ <i>antwoord</i>				(2)
1.5	Mean will increase by R200 <i>Gemiddelde sal vermeerder met R200</i> Median will increase by R200 <i>Mediaan sal vermeerder met R200</i> Standard deviation will remain the same <i>Standaardafwyking sal dieselfde bly</i>					✓ increase by R200 <i>vermeerder met R200</i> ✓ increase by R200 <i>vermeerder met R200</i> ✓ remain the same <i>bly dieselfde</i>				(3)
										[11]

QUESTION 2/VRAAG 2



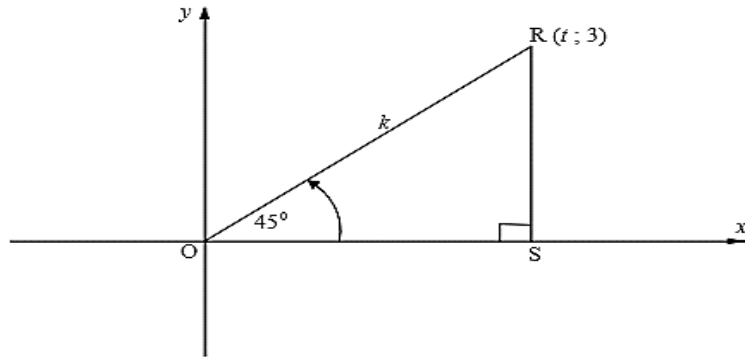
2.1	140 employees/werknemers	✓ answer/antwoord	(1)
2.2	$30 < x \leq 40$	✓ answer/antwoord	(1)
2.3	Reading from the graph / lees vanaf grafiek 32 employees/werknemers	✓ Reading/lees ✓ answer/antwoord	(2)
2.4	$118 - 24 = 94$	✓ $118 - 24$ ✓ answer/antwoord	(2)
2.5	25 th percentile is at 35 = R25 25 ^{ste} persentiel is by 35 = R25 75 th percentile is at 105 = R38 75 ^{ste} persentiel is by 105 = R38	✓ Q_1 ✓ Q_3	(2)
2.6	$IQR/IKW = Q_3 - Q_1 = R38 - R25$ $= R13$	✓ method/metode ✓ answer/antwoord	(2)
			[10]

A coordinate plane with x and y axes. The origin is labeled O. A point P is located at (0, 2). A line segment PQ is drawn, where Q is at (3, 0). A line segment PR is drawn, where R is at (1, 6). A parallelogram is constructed by drawing a line through Q parallel to PR and a line through R parallel to PQ. These lines intersect at point S(13, 3). The vertices of the parallelogram are P, Q, R, and S. The sides of the parallelogram are PQ, QR, RS, and SP. The triangle PQR is shaded. The coordinates of the vertices are: P(0 ; 2), Q(3 ; 0), R(1 ; 6), and S(13 ; 3).

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3.5	$M_{WS}\left(\frac{13+1}{2}; \frac{3+6}{2}\right)$ $M_{WS}\left(7; \frac{9}{2}\right)$ $M_{QV}\left(\frac{11+3}{2}; \frac{7+2}{2}\right)$ $M_{QV}\left(7; \frac{9}{2}\right)$ $\therefore$ WVSQ is a parm (Diag. bisect each other.) WVSQ is 'n $\parallel^m$ (Hoeklyne halveer mekaar)	✓ correct substitution <i>korrekte vervanging</i> ✓ coordinates of midpt of WS <i>koördinate van midpt van WS</i> ✓ coordinates of midpt of QV <i>koördinate van midpt van QV</i> ✓ reason/rede	(4)
3.6	$WV = \sqrt{(1-11)^2 + (6-7)^2}$ $= \sqrt{101}$	✓ correct substitution <i>korrekte vervanging</i> ✓ answer/antwoord	(2)
3.7	Let inclination of WV to be θ and QU be α <i>Laat inklinasie van WV, θ wees en van QU, α wees</i> $\tan \theta = m_{WV} = \frac{1}{10}$ $\theta = 5,71^\circ$ $\tan \alpha = m_{QU} = \frac{3}{2}$ $\alpha = 56,31^\circ$ $\therefore \hat{WUQ} = \hat{UQS} = 50,6^\circ$ [alt $\angle$ s, $WV \parallel QS$] [verv. $\angle$ e, $WV \parallel QS$]	✓ $\tan \theta = m_{WV}$ ✓ size of θ / grootte van θ ✓ size of α / grootte van α ✓ answer/antwoord	(4)
3.8	$WU = \frac{\sqrt{101}}{2}$ $UQ = \sqrt{(6-3)^2 + \left(\frac{13}{2}-2\right)^2} = \frac{3\sqrt{13}}{2}$ <i>A. of ΔQUW</i> $= \frac{1}{2} \times \frac{3\sqrt{13}}{2} \times \frac{\sqrt{101}}{2} \times \sin(50,60^\circ)$ $= 10,50$ $\frac{\text{Area of PQRO}}{\text{Area of } \Delta QUW} = \frac{2 \times 3}{10,50}$ $= \frac{4}{7}$	✓ WU ✓ UQ ✓ correct subst. in to Area rule <i>korrekte verv. in oppervl. reël</i> ✓ A. of Δ . / Oppervl. van Δ ✓ correct subst. in A. of rect. <i>korrekte verv. in Opp. van regh</i> ✓ answer/antwoord	(6)
			[27]

QUESTION 4 / VRAAG 4

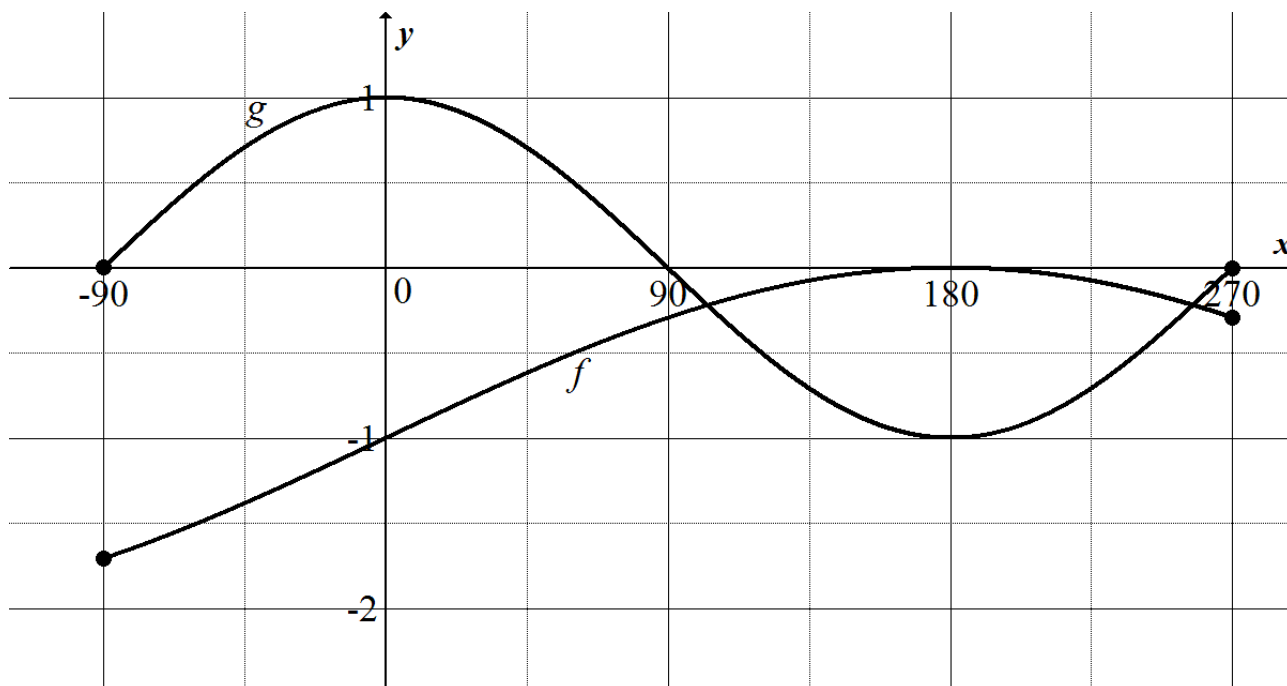


4.1.1	$\tan 45^\circ = \frac{3}{t}$ $t = 3$	✓ correct trig ratio ✓ korrekte trig. verhouding ✓ t-value/waarde	(2)
4.1.2	$\sin 45^\circ = \frac{3}{k}$ OR / OF $\cos 45^\circ = \frac{t}{k} = \frac{3}{k}$ $k = 3\sqrt{2}$ $k = 3\sqrt{2}$	✓ correct trig ratio ✓ korrekte trig. verhouding ✓ k-value/waarde	(2)
4.2.1	$8\cos^2 \alpha - 2 = 0$ $\cos^2 \alpha = \frac{1}{4}$ $\cos \alpha = \frac{1}{2}$	✓ simplification ✓ vereenvoudiging ✓ answer/antwoord	(2)
4.2.2	 OR / OF $y = \sqrt{2^2 - 1^2}$ <i>Pyth</i> $= \sqrt{3}$ $4\left[\tan^2 \alpha - \cos^2 \alpha\right] = 4\left[\left(\frac{\sqrt{3}}{1}\right)^2 - \left(\frac{1}{2}\right)^2\right]$ $= 11$	✓ diagram/diagram OR / OF ✓ $y = \sqrt{3}$ ✓ correct substitution ✓ korrekte vervanging ✓ answer/antwoord	(3)

4.3.1	$\frac{\cos x \cdot \tan x}{2[\sin x \cdot \cos(x - 90^\circ) + \cos x \cdot \cos(-x)]}$ $= \frac{\cos x \cdot \frac{\sin x}{\cos x}}{2[\sin x \cdot (\sin x) + \cos x \cdot (\cos x)]}$ $= \frac{\sin x}{2[\sin^2 x + \cos^2 x]}$ $= \frac{\sin x}{2(1)}$ $= \frac{\sin x}{2}$	$\checkmark \frac{\sin x}{\cos x}$ $\checkmark \sin x \quad \checkmark \cos x$ $\checkmark$ simplification <i>vereenvoudiging</i> $\checkmark \sin^2 x + \cos^2 x = 1$ $\checkmark$ answer/antwoord	(6)
4.3.2	Period/Periode = 360°	$\checkmark$ answer/antwoord	(1)
4.4	$\frac{\cos 42^\circ \sin 48^\circ - \tan^2(-45^\circ)}{\cos^2 132^\circ}$ $= \frac{\cos 42^\circ \cdot \cos 42^\circ - (-\tan 45^\circ)^2}{(\cos(90^\circ + 42^\circ))^2}$ $= \frac{\cos^2 42^\circ - (-1)^2}{(-\sin 42^\circ)^2}$ $= \frac{\cos^2 42^\circ - 1}{\sin^2 42^\circ}$ $= \frac{1 - \sin^2 42^\circ - 1}{\sin^2 42^\circ}$ $= -1$	$\checkmark \sin 48^\circ = \cos 42^\circ$ $\checkmark (-\tan 45^\circ)^2$ $\checkmark (-\sin 42^\circ)^2$ $\checkmark \cos^2 42^\circ = 1 - \sin^2 42^\circ$ $\checkmark$ answer/antwoord	(5)
4.5.1	$\text{LHS/LK} = \tan \beta - \sin \beta \cdot \cos \beta$ $= \frac{\sin \beta}{\cos \beta} - \sin \beta \cdot \cos \beta$ $= \frac{\sin \beta - \sin \beta \cos^2 \beta}{\cos \beta}$ $= \frac{\sin \beta (1 - \cos^2 \beta)}{\cos \beta}$ $= \tan \beta \cdot \sin^2 \beta$ $= \text{RHS / RK}$	$\checkmark \frac{\sin \beta}{\cos \beta}$ $\checkmark$ simplification <i>vereenvoudiging</i> $\checkmark$ common factor <i>gemeenskaplike faktor</i>	(3)
4.5.2	$\beta = -90^\circ$ and / en $\beta = 90^\circ$	$\checkmark \beta = -90^\circ \quad \checkmark \beta = 90^\circ$	(2)

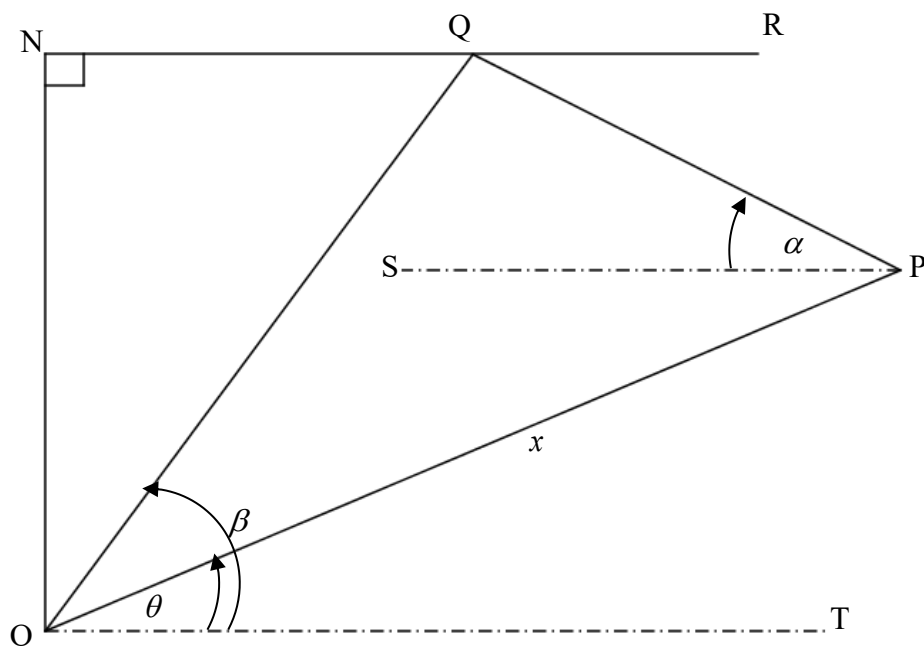
4.6	$3 \sin \theta = -2 \cos^2 \theta$ $2(1 - \sin^2 \theta) + 3 \sin \theta = 0$ $2 - 2 \sin^2 \theta + 3 \sin \theta = 0$ $2 \sin^2 \theta - 3 \sin \theta - 2 = 0$ $(2 \sin \theta + 1)(\sin \theta - 2) = 0$ $\sin \theta = -\frac{1}{2}$ or / of $\sin \theta \neq 2$ $\theta = 210^\circ + 360.k \quad \theta = 330^\circ + 360.k, \quad k \in \mathbb{Z}$	$\checkmark 1 - \sin^2 \theta$ $\checkmark$ standard form <i>standaardvorm</i> $\checkmark$ factors/faktore $\checkmark$ both equations <i>beide vergelykings</i> $\checkmark \theta = 210^\circ + 360.k$ $\checkmark \theta = 330^\circ + 360.k, \quad k \in \mathbb{Z}$	(6)
			[32]

QUESTION 5/VRAAG 5



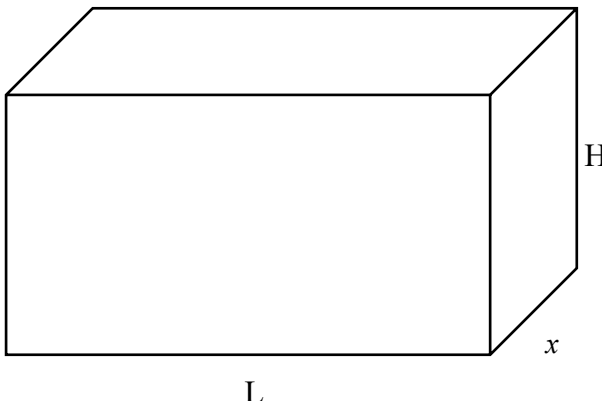
5.1	$a = 1$ $q = 0$	✓ value of a / waarde van a ✓ value of q / waarde van q	(2)
5.2	$y \in [-2; 0]$ or / of $-2 \leq y \leq 0$	✓ correct critical value korrekte kritieke waarde ✓ correct notation korrekte notasie	(2)
5.3	Amplitude = 1	✓ correct answer korrekte antwoord	(1)
5.4	$x = 0^\circ$	✓ answer / antwoord	(1)
5.5	$-90^\circ \leq x \leq 90^\circ$	✓✓ correct answer korrekte antwoord	(2)
5.6	$h(x) = g(x + 90^\circ)$ $= \cos(x + 90^\circ)$ $= -\sin x$	✓ $\cos(x + 90^\circ)$ ✓ $-\sin x$	(2)
			[10]

QUESTION 6/VRAAG 6

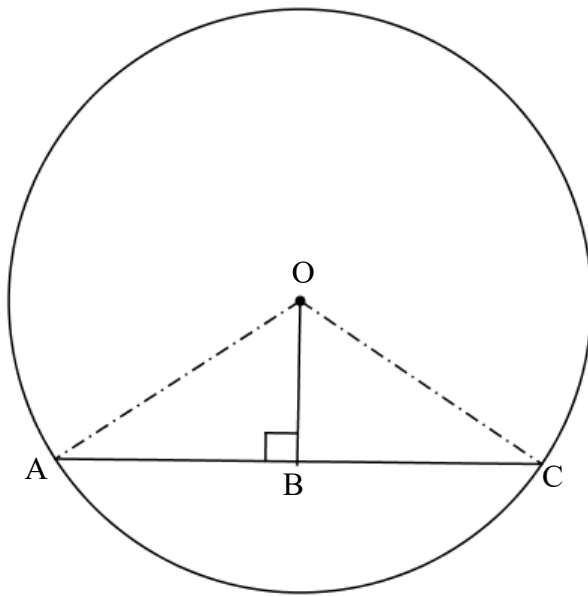


6.1	$\hat{SPO} = \theta + \alpha$ [alt. $\angle$ s, verw. $\angle$ e : $SP \parallel OT$] $\therefore$ $\hat{QOP} = \beta - \theta$ $\hat{OQP} = 180^\circ - (\theta + \alpha) - (\beta - \theta)$ [$\angle$ s in Δ / $\angle$ e in Δ] $= 180^\circ - (\beta + \alpha)$	$\checkmark \hat{SPO} = \theta + \alpha$ $\checkmark$ using sum of angles in Δ <i>gebruik van som van $\angle$e in Δ</i>	(2)
6.2	$\frac{OQ}{\sin \hat{P}} = \frac{OP}{\sin \hat{Q}}$ $\frac{OQ}{\sin(\alpha + \theta)} = \frac{x}{\sin[180^\circ - (\beta + \alpha)]}$ $OQ = \frac{x \cdot \sin(\alpha + \theta)}{\sin(\beta + \alpha)}$	$\checkmark$ application of sine rule <i>toepassing van sinusreël</i> $\checkmark$ correct substitution <i>korrekte vervanging</i> $\checkmark$ reduction $\sin(\beta + \alpha)$ <i>reduksie $\sin(\beta + \alpha)$</i>	(3)
6.3	$\frac{5\sqrt{2}}{\sin 45^\circ} = \frac{x}{\sin 90^\circ}$ $x = 10$	$\checkmark$ correct substitution <i>korrekte vervanging</i> $\checkmark$ answer/antwoord	(2)
			[7]

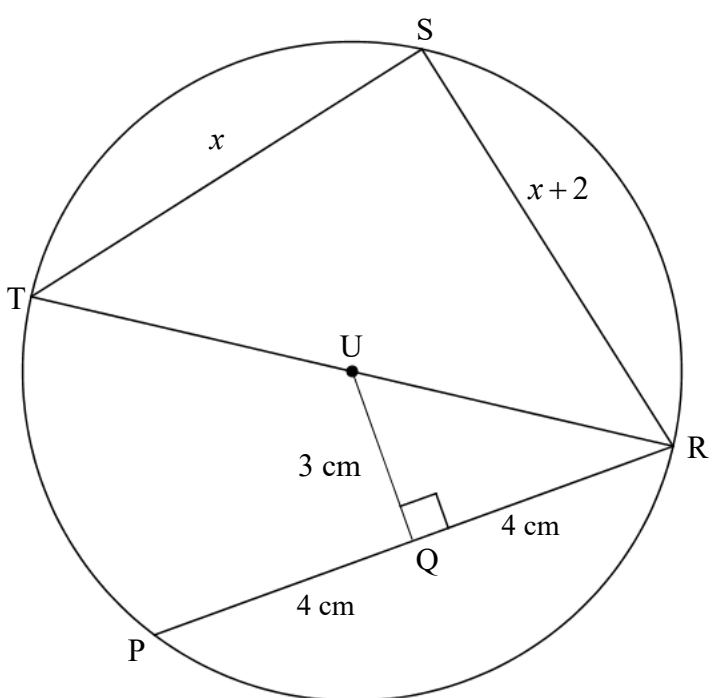
QUESTION 7/VRAAG 7

			
7.1	$L = 2x$ $H = 3 + x$	✓ $L = 2x$ ✓ $H = 3 + x$	(2)
7.2	$SA = 2(2x \times x) + 2(x(3 + x)) + 2(2x(3 + x))$ $= 4x^2 + 6x + 2x^2 + 12x + 4x^2$ $= 10x^2 + 18x$	✓ correct substitution / korrekte vervanging ✓ simplification / vereenvoudiging ✓ answer / antwoord	(3)
7.3	$10x^2 + 18x = 784$ $10x^2 + 18x - 784 = 0$ $5x^2 + 9x + 392 = 0$ $(5x + 49)(x - 8) = 0$ $x \neq \frac{40}{5} \text{ or } x = 8$	✓ correct substitution/ korrekte vervanging ✓ standard form/standaardvorm ✓ factors/faktore ✓ correct answer/korrekte antwoord	(4)
7.4	$V = (8)(2 \times 8)(3 + 8)$ $= 1408$	✓ correct substitution into volume / korrekte vervanging in volume ✓ answer/antwoord	(2)
			[11]

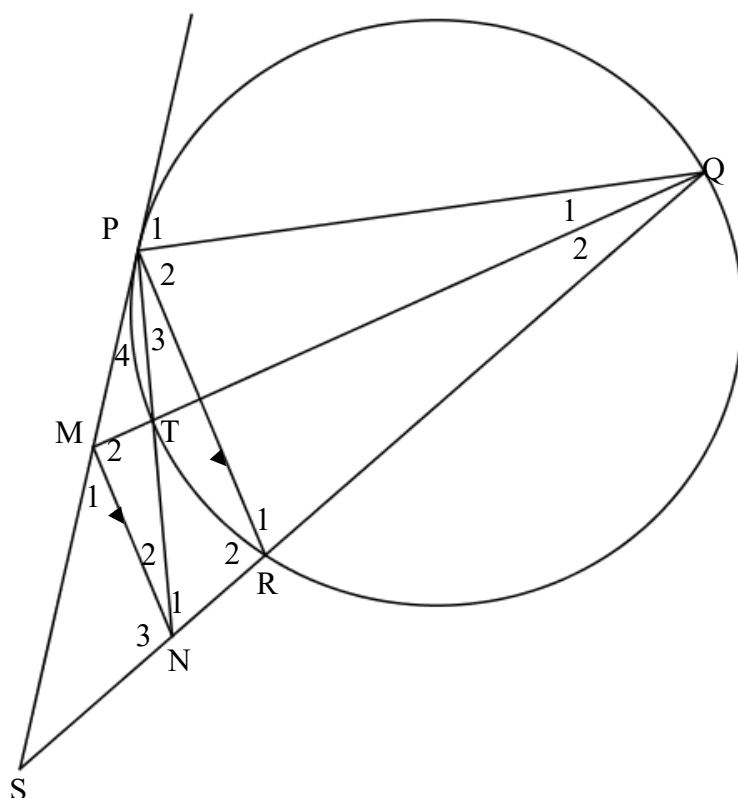
QUESTION 8/VRAAG 8



8.1	Constructions: Join OA and OC <i>Konstruksies: Verbind OA en OC</i> Proof in $\triangle AOB$ and $\triangle COB$ <i>Bewys in $\triangle AOB$ en $\triangle COB$</i> 1. $AO = CO$ [radii]/[radiusse] 2. $\angle ABO = \angle CBO = 90^\circ$ [$\angle$s on a str. line]/[$\angle$op reguitlyn] 3. $OB = OB$ [common]/[gemeen] $\therefore \triangle AOB \equiv \triangle COB$ [RHS]/[90° Sk S] $\therefore AB = BC$	✓ construction <i>konstruksie</i> ✓ S/R ✓ S/R ✓ S ✓ R	(5)
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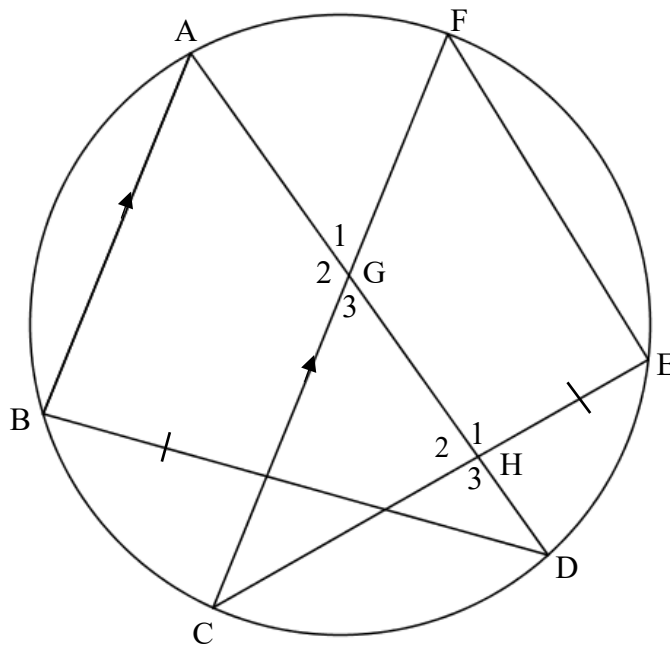
8.2			
8.2.1	$PQ = QR = 4 \text{ cm}$ [line from centre $\perp$ to chord] [lyn vanaf middelpunt $\perp$ op koord] $UR = \sqrt{4^2 + 3^2}$ [Pyth.theo]/ [Stelling van Pythagoras] $= 5$	$\checkmark$ S $\checkmark$ R $\checkmark$ correct substitution/ korrekte vervanging $\checkmark$ answer/antwoord	(4)
8.2.2	$TR = 2(UR) = 10$ [diameter]/ [middellyn] $\hat{S} = 90^\circ$ [$\angle$ in semi – circle]/ [$\angle$ in semi – sirkel] $10^2 = x^2 + (x + 2)^2$ [Pyth. theo]/ [Stelling van Pythagoras] $100 = x^2 + x^2 + 4x + 4$ $2x^2 + 4x - 96 = 0$ $x^2 + 2x - 48 = 0$ $(x - 6)(x + 8) = 0$ $x = 6 \text{ or / of } x \neq -8$ $\therefore TS = 6$	$\checkmark$ S $\checkmark$ R $\checkmark$ correct substitution/ korrekte vervanging $\checkmark$ standard form / standaardvorm $\checkmark$ factors / faktore $\checkmark$ correct value of x / korrekte waarde van x	(6)
			[15]

9.2



9.2.1	$\hat{P}_1 = \hat{R}_1$ [tan chord theo] / [raaklyn – koord stelling] $\hat{P}_1 = \hat{M}\hat{N}Q$ [corresp. $\angle$ s, / ooreenk. $\angle$ e : $MN \parallel PR$] $\hat{R}_1 = \hat{P}\hat{T}Q$ [$\angle$ s same seg] / [$\angle$ e in dies. segment]	$\checkmark$ S $\checkmark$ R $\checkmark$ S $\checkmark$ R Any 2 angles with reasons / Enige 2 hoeke met redes	(4)
9.2.2	$\hat{S}\hat{P}Q = \hat{Q}$ [tan chord theo] / [raaklyn – koord stelling] $\hat{S}\hat{P}Q = \hat{M}_1$ [corresp. $\angle$ s, / ooreenk. $\angle$ e : $MN \parallel PR$] $\therefore \hat{Q}_1 = \hat{M}_1$ $\therefore MNQP$ is a cyclic quad [conv. ext. $\angle$ of a cyclic quad] <i>MNQP is 'n koordevh. [omgek. buite $\angle$ van koordevh]</i>	$\checkmark$ S $\checkmark$ R $\checkmark$ S $\checkmark$ R	(4)
9.2.3	$\hat{P}_3 = \hat{N}_2$ [alt. $\angle$ s, $MN \parallel PR$] / [verw. $\angle$ e : $MN \parallel PR$] $\hat{N}_2 = \hat{Q}_1$ [$\angle$ s same seg] / [$\angle$ e in dies. segment] $\hat{Q}_1 = \hat{P}_4$ [tan chord theo] / [raaklyn – koord stelling] $\therefore \hat{P}_3 = \hat{P}_4$ <i>PN bisects $\hat{S}\hat{P}R$ / PN halveer $\hat{S}\hat{P}R$</i>	$\checkmark$ S/R $\checkmark$ S $\checkmark$ R $\checkmark$ S/R	(4)
			[22]

QUESTION 10/VRAAG 10



$\hat{G}_3 = \hat{A}$	[corresp. $\angle$ s / ooreenk. $\angle$ e, $GC \parallel AB$]	✓ S	✓ R	
$\hat{A} = \hat{F}$	[$\angle$ s subt by = chords] / [$\angle$ e onderspan deur = koorde]	✓ S	✓ R	
$\therefore \hat{G}_3 = \hat{F}$		✓ R		
$\therefore AB \parallel FE$	[corresp. $\angle$ s, =] / [ooreenkomstige $\angle$ e =]			[5]

TOTAL/TOTAAL: 150